



9/1/2026

# N + - Network Fundamentals & Building Networks.

Build a SOHO Network.

## TASK 1.

- ❖ Draw a network diagram for a small office/home office (SOHO) setup that includes: one broadband internet connection, one Wi-Fi router, two laptops, one smart TV, and one printer. Clearly label each device and connection type (wired or wireless).

### ➤ SOHO Network Setup:

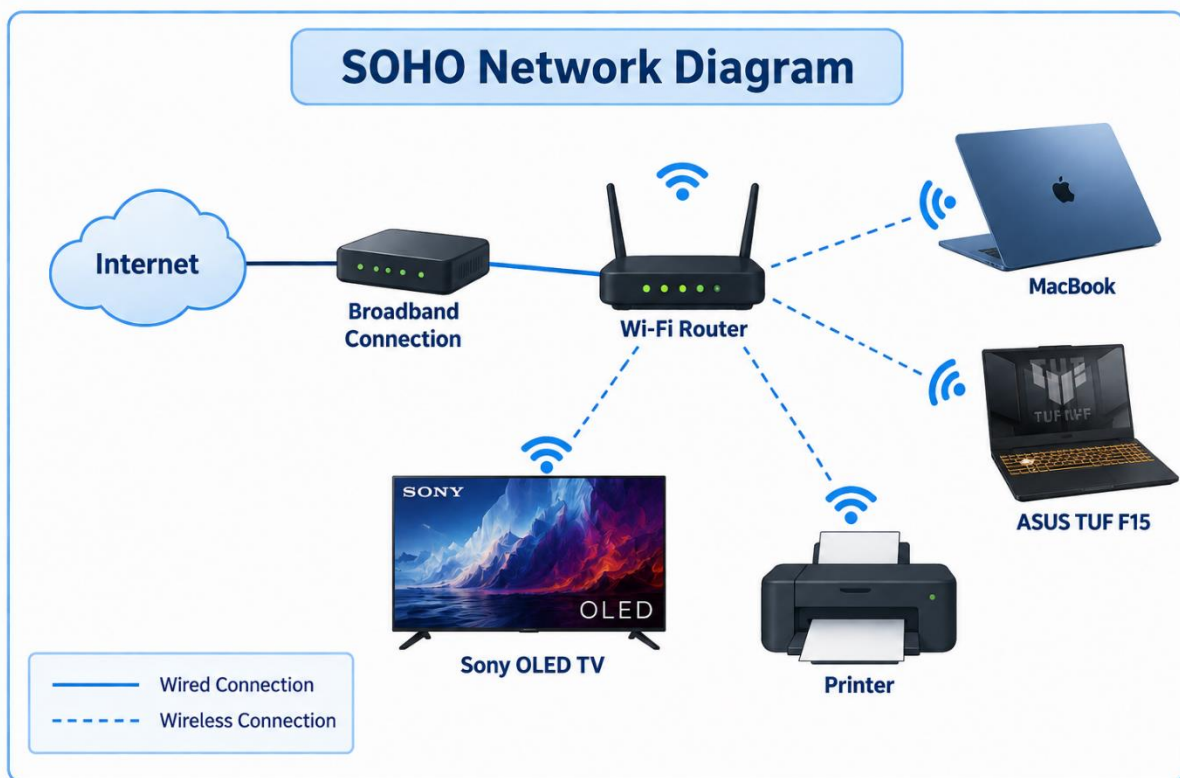


Figure : SOHO Network Diagram.

### Explanation:

- ✓ The diagram shows a **small office/home office (SOHO) network** where the broadband Internet connection is connected to a **Wi-Fi router**. The router provides wireless connectivity to **two laptops, a Sony OLED TV, and a printer**. **Solid lines represent wired connections**, while **dashed lines represent wireless connections**.

## TASK 2.

❖ List all the hardware components (with brand/model suggestions) you would need to physically build the SOHO network diagrammed above, including cables and any switches or extenders if needed, Check current listings on Flipkart or Amazon for typical device models used in Indian homes/offices.

➤ Hardware Components for SOHO Network:

| Component           | Suggested Model        | Use                                 |
|---------------------|------------------------|-------------------------------------|
| Broadband Modem/ONT | ISP-provided           | Provides Internet connection        |
| Wi-Fi Router        | TP-Link Archer C6      | Connects the devices to the network |
| Laptops             | MacBook & ASUS TUF F15 | Connect wirelessly to the router    |
| Smart TV            | Sony OLED TV           | Connects through Wi-Fi              |
| Printer             | Wi-Fi Printer          | Used for wireless printing          |
| Ethernet Cable      | Cat6 RJ45              | Used for wired connections          |
| Switch              | TP-Link TL-SG1008D     | Optional, for extra wired devices   |
| Wi-Fi Extender      | TP-Link TL-WA850RE     | Optional, if Wi-Fi coverage is weak |

## TASK 3.

❖ Write step-by-step instructions to configure the Wi-Fi router for your SOHO network, including setting up a unique SSID, enabling WPA2 security, and changing the default admin password.

### ➤ Wi-Fi Router Configuration:

#### Steps:

- Connect the computer to the **Wi-Fi router** and open its configuration page.
- Go to **Wireless Settings** and change the default SSID to a unique name, such as **Rishi\_HomeWiFi**.
- Open **Wireless Security** and select **WPA2-PSK** as the security mode.
- Set a strong **Wi-Fi password** and save the settings.
- Go to **Administration/Management** settings and change the default **admin username/password**.
- Save the changes and restart the router if required.

**Result:** The router is configured with a unique SSID, **WPA2 security**, and a changed administrator password.

# TASK 4.

❖ **Configure IP addresses for all devices in your SOHO network: assign static IPs to the printer and smart TV, and use DHCP for the laptops. Show a table listing each device, its MAC address (make one up), and the assigned IP address.**

➤ **IP Address Configuration:**

| Device        | MAC Address       | IP Address    | Configuration |
|---------------|-------------------|---------------|---------------|
| MacBook       | 00:1A:2B:3C:4D:01 | 192.168.10.10 | DHCP          |
| ASUS TUF F15  | 00:1A:2B:3C:4D:02 | 192.168.10.11 | DHCP          |
| Sony OLED TV  | 00:1A:2B:3C:4D:03 | 192.168.10.20 | Static        |
| Wi-Fi Printer | 00:1A:2B:3C:4D:04 | 192.168.10.21 | Static        |

The laptops receive their IP addresses automatically using **DHCP**, while the **printer and Sony OLED TV** use fixed static IP addresses.

## TASK 5.

- ❖ Explain how you would troubleshoot if one of the laptops cannot connect to the internet in your SOHO network. List at least 3 possible causes and what steps you would take to diagnose and fix each issue.

➤ Troubleshooting Internet Connectivity:

| Possible Cause                | How to Diagnose                                                              | How to Fix                                                    |
|-------------------------------|------------------------------------------------------------------------------|---------------------------------------------------------------|
| 1. Weak Wi-Fi signal          | Check the Wi-Fi signal strength on the laptop and move closer to the router. | Move the laptop closer to the router or use a Wi-Fi extender. |
| 2. Incorrect IP configuration | Check the laptop's IP address and gateway using ipconfig.                    | Set the laptop to DHCP and reconnect to the Wi-Fi network.    |
| 3. Router/Internet problem    | Try accessing the Internet from another device connected to the same router. | Restart the router and check the broadband connection.        |

**Result:** These steps help identify whether the problem is with the **laptop, Wi-Fi connection, or router/Internet service**.